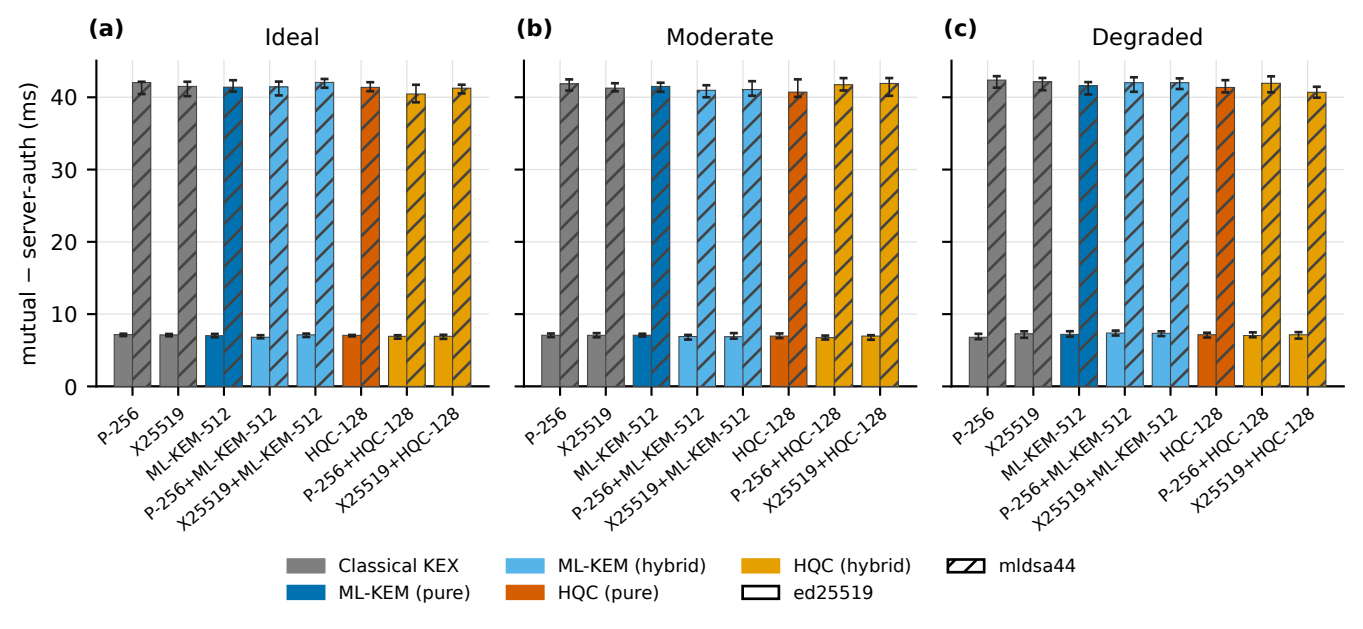
(a)
Ideal
(b)
Moderate
(c)
Degraded
mutual − server-auth (ms)
0
10
20
30
40
P-256
X25519
ML-KEM-512
P-256+ML-KEM-512
X25519+ML-KEM-512
HQC-128
P-256+HQC-128
X25519+HQC-128
Classical KEX
ML-KEM (pure)
ML-KEM (hybrid)
HQC (pure)
HQC (hybrid)
ed25519
mldsa44